

Unit -2: FRAMEWORK AND MODELS

AI Governance by Human-right centred design, Normative models, Role of professional norms, Teaching Machines to be Moral.

1. AI Governance by Human-Right Centred Design:

The number and variety of topics in this volume illustrate the width, diversity of content and vagueness of the boundaries of ‘AI Ethics’ as a domain of inquiry. Within this discourse, increasing attention has been drawn to the capacity of socio-technical systems that utilise data-driven algorithms to classify, to make decisions and to control complex systems, including the use of machine learning and large datasets to generate predictions about future behaviour (hereafter ‘AI’ systems¹) in ways that may interfere with human rights. The recent Cambridge Analytica scandal revealed how unlawfully harvested Facebook data from millions of voters in the UK, the US and elsewhere enabled malign actors to engage in political microtargeting through the use of AI-driven social media content distribution systems, thereby interfering with their right to free and fair elections and thus threatening the integrity of democratic processes. The increasing use of algorithmic decision-making (‘ADM’) systems to inform custodial and other decisions within the criminal justice process may threaten several human rights, including the right to a fair trial, the presumption of innocence and the right to liberty and security. Systems of this kind are now used to inform, and often to automate, decisions about an individual’s eligibility and entitlement to various benefits and opportunities, including housing, social security, finance, employment and other life-affecting opportunities, potentially interfering with rights of due process and rights to freedom from unfair or unlawful discrimination.

1.1 Why should human rights lie at the core of ‘AI ethics’?

Within contemporary discussions of ‘AI ethics’, there is no agreed set of ethical standards that should govern the operation of AI, reflected in the variety of ethical standards espoused in various voluntary ‘AI ethics codes’ that have emerged in recent years. The salience of ‘AI ethics’ reflects welcome recognition by the tech industry and policy-makers that AI systems may have significant adverse impacts, with some values commonly appearing in these discussions, particularly those of ‘transparency’,

‘fairness’ and ‘explainability’. Yet the vagueness and elasticity of the scope and content of ‘AI ethics’ has meant that it currently operates as an empty vessel into which anyone (including the tech industry, and the so-called Digital Titans) can pour their preferred ‘ethical’ content. Without an agreed framework of norms that clearly identifies and articulates the relevant ethical standards which AI systems should be expected to comply with, little real progress will be made towards ensuring that these systems are in practice designed, developed and deployed in ways that will meet widely accepted ethical standards. Although there is scope for reasonable disagreement concerning what ‘ethical conduct’ requires in any given case, a core set of agreed norms that constitute the basic minimum below which conduct cannot fall if it can be appropriately characterised as ethically acceptable must be identified.

1.2 Why ‘human rights-centred design, deliberation and oversight’ of AI?

The ineffectiveness of the prevailing self-regulatory approach to ‘ethical AI’ demonstrates that an alternative governance model is needed: one that is (1) anchored in human rights norms and a human rights approach (2) utilises a coherent and integrated suite of technical, organisational and evaluation tools and techniques, that is (3) subject to legally mandated external oversight by an independent regulator with appropriate investigatory and enforcement powers, and (4) provides opportunities for meaningful stakeholder and public consultation and deliberation. In the next section, we develop a governance framework for AI intended to do just that, which we call ‘human rights-centred design, deliberation and oversight’. Although much more foundational work remains to be done, both to specify the content and contours of this approach more fully and to render it capable of practical implementation, we believe that our proposed framework offers a concrete approach that can bring an end to ‘ethics washing’ by securing the design, development and deployment of human rights-compliant AI systems in real world settings.

1.3 Core Principles of Human Rights-Centred Design, Deliberation and Oversight

The methods and techniques listed above vary widely in their disciplinary foundations and in the original contexts of their development. Our proposal seeks to draw them together in an integrated manner, appropriately adapted towards ensuring conformity with human rights norms, as the basis for a comprehensive design and governance regime constructed around the following four core principles in which human rights norms provide the foundational ethical standards which AI systems must demonstrably comply with:

1. Design and deliberation

2. Assessment, testing and evaluation
3. Independent oversight, investigation and sanction
4. Traceability, evidence and proof.

Each of these principles is briefly outlined below.

Principle 1:

Design and deliberation Central to our approach is a requirement that AI systems should be designed and configured to operate in ways that are compliant with universal human rights standards (such as those, for example, set out in the ECHR), and that, at least for systems identified during the design and development phase as posing a ‘high risk’ of interfering with human rights, affected stakeholders are consulted about the proposal and given opportunities to express their views about the proposed system’s potential impact, in discussion with the system’s designers. Consultation with affected stakeholders and the general public during the initial phases of system design contributes to the overall legitimacy of the regime, understood in terms of respect for democratic values and affected communities, and should help system designers to identify which aspects of the system’s design and proposed operation need reconsideration. Where the risks to human rights are assessed as ‘high’ or ‘very high’³³ this would trigger an obligation on system designers to reconsider and redesign the system and/or proposed business model³⁴ in order to reduce those risks to a form and level regarded as tolerable (understood in terms of a human rights approach to the resolution of conflict between rights and collective interests), in ways that duly accommodate concerns expressed by affected.

Principle 2:

Assessment, testing and evaluation Users and others affected by the operation of AI systems (including the general public) can only have justified confidence that AI systems do in fact comply with human rights standards if these systems can be subjected to formal assessment and testing to evaluate their compliance with human rights standards, and if these occur regularly throughout the entire lifecycle of system development: from the initial formulation of a proposal through to design, specification, development, prototyping, and real world implementation, and which includes periodic evaluation of the data sets upon which the system has been trained and upon which it operates.

Principle 3:

Independent oversight, investigation and sanction In order to provide meaningful assurance that AI systems are in fact human rights compliant, rather than merely claiming to be human rights-compliant, independent oversight by an external, properly resourced, technically competent oversight body invested with legal powers of investigation and sanction is essential. 39 Because the operation of market-forces cannot provide those who design, develop and deploy AI systems with sufficient incentives to invest the required resources necessary to ensure that AI systems are human rights compliant, our proposed approach must operate within a legally mandated institutional structure, including an oversight body with a duty to monitor and enforce substantive and procedural (regulatory) requirements, including those concerning robust design, verification, testing, and evaluation (including appropriate documentation demonstrating that these requirements have been fulfilled) supported by legally mandated stakeholder and public consultation where proposed AI systems pose a ‘high risk’ to human rights.

Principle 4:

Traceability, evidence and proof In order to facilitate meaningful independent oversight and evaluation, AI systems must be designed and built to secure auditability: this means more than merely securing transparency, but is aimed at ensuring that they can be subject to meaningful review, thus providing a concrete evidential trail for securing human accountability over AI systems. Not only is it necessary that systems be constructed to produce evidence that they operate as desired, 46 there must be a legal obligation to do so, requiring that crucial design decisions, the testing/assessment process and the outcome of those processes, and the operation of the system itself, are properly documented and provide a clear evidence trail that can be audited by external experts. Drawing again on the experience of the ‘safety case’ approach, which entails imposing a legal duty on operators to demonstrate to the regulator that robust and comprehensive systems are in place that reduce safety risks to a level that is ‘as low as reasonably practical’, we envisage the imposition of a suitably formulated legal duty on AI systems developers, owners and operators to demonstrate that these systems are human rights compliant

2. Normative Model

In this paper, I provide a normative model of explanation for the output of AI algorithms used in legal practice. I focus on binary classification algorithms due to their extensive use in the field. In the last part of the paper, I examine the model’s compatibility with causal explanations provided by Answer Set Programming (ASP)

causal models. The motivation for proposing this model is the necessity for providing explanations for the output of legal AI. From the multiplicity of arguments supporting that necessity, the proposed model addresses the argument that legal AI's output should be objectionable. That can be achieved only if the explanation of the output has a form that makes it amenable to evaluation by legal practitioners. Hence, I firstly provide a normative model for the explanations used by legal practitioners in their practice (CLMLP) and then I provide the normative model for the explanations of legal AI's outputs (EXPBC) that I base on CLPLP. CLPLP in its turn is based on the Classical Model of Science (CMS) which is the normative model of explanations that every "proper" science should follow according to philosophers throughout history.

2.1 A normative model of legal practice:

2.1.1 The Classical Model of Science 2.1.

Legal practice Before constructing a normative model of legal practice, I have to clarify what legal practice is in the context of this paper. By "legal practice", I mean the totality of the activities of the legal experts that are authorised to take part in the process of deciding a judgement (e.g., the lawyers of the defense and the prosecution, the judges of the authorised court). Those activities can be for instance the arguments the defence makes during a trial and all of the defence's preparatory work before the trial. The authorisation of the legal experts is stipulated by law. E.g., the European Convention of Human Rights (the Convention) stipulates that the authorised court for deciding whether there has been a violation of the Convention's articles is the European Court of Human Rights (ECtHR). I borrow this construal of legal practice from . My motivation is that according to this construal, the produced results of legal practice are legally binding. And being legally binding will be a necessary property for the conception of truth I will employ in the proposed normative model of legal practice.

2.1.2 The Classical Model of Science (CMS)

The normative model of legal practice I will provide will be based on the Classical Model of Science (CMS). CMS is a posterior reconstruction of how the ideal scientific explanation for every "proper science" should look like according to philosophers throughout history. One may object to my decision to provide a normative model of legal practice using the CMS since the latter is about sciences and not about a practice. However, even if legal practice is not a science - whatever that means - my arguments supporting that legal practice fits into the CMS will still hold since none of those arguments is based on the premises that legal practice is a science. Let's move to

the presentation of the CMS. I will provide a less detailed account of the CMS which I will call CMSsum. For the construction of EXPBC, there is no need to argue how legal practice fits into the more detailed version of CMSsum found. Specifically, EXPBC will be a minimal model. By “minimal”, I mean that one can further expand and/or precisify parameters of that model using the more detailed version of the CMSsum. However, for that to be done properly, I would have had to exceed the word limit.

2.1.3. Applying CMS sum to legal practice

As noted by, for each application of the CMS, CMS’s parameters may have to be modified to fit the particularities of that application. Below, I perform such modifications to apply CMSsum to legal practice. In §2.4, we sum up the outcome of those modifications to introduce a normative model of legal practice which I will name CMSLP. Applying (SUM.3): Legal reasoning can be construed as rule-based reasoning. One way to perform such a construal is to view a legal inference as an inference whose premises consist of particular facts and general rules like the following example borrowed.

A particular fact is a proposition while a (general) rule is a norm. Intuitively, the difference is that propositions describe which state of affairs is the case (e.g., “Alice is paying taxes.”) while norms stipulate which state of affairs should be the case (e.g., “Alice should pay taxes.”). Since legal inferences - like Example 1 - contain both propositions and norms, we cannot explain the inferences’ conclusions without appealing to the referred norms. Consequently, CMSLP has to contain norms apart from propositions. Moreover, Example 1’s conclusion is also a rule - not a general, but a particular one - and hence a norm. In other words, we can derive new norms from other norms and propositions. Consequently, the hierarchy *HP* in CMSLP contains both norms and propositions which are ordered based on derivation relations; ’s constituents situated higher are derived from those situated lower. E.g., norm *n2* is situated higher in *HP* than norm *n1* and proposition *p1*. The derivation relations whose conclusions are norms are those of subsumptive-deductive inferences: the particular fact is subsumed by rule and via deduction the conclusion.

A normative model of the explain ability of upload filters 3.1.

The model Assume a set of norms $\mathcal{N}$ that are legally binding for a set of data D (e.g., hateful tweets). Then, based on the norms, D is segmented into two disjoint sets: $D = D\mathcal{N} \cup \bar{D}\mathcal{N}$. Data d are elements of $D\mathcal{N}$ if and only if they do not violate the norms of $\mathcal{N}$. Equivalently, d are elements of $\bar{D}\mathcal{N}$ if and only if they do violate the norms of $\mathcal{N}$.

Since $\mathcal{N}$ are norms that are legally binding for D , any judgement on the violation of $\mathcal{N}$ by D 's data is true if and only if it coincides with the judgments of the authorised legal practitioners. Since I want to provide a normative model and not a descriptive one, I assume that there is an algorithm $\mathcal{F}$ that segments (or filters) D to the two disjoint sets $D = D\mathcal{N} \cup D\bar{\mathcal{N}}$ with 100% accuracy (ideal - normative case). Propositions of the form $d \in D\mathcal{N}$ - where $d \in \{D\}$ - are judgements for the legality of d . Since p represents a judgement, we can substitute it by a norm np that represents the same judgement. Since np is a judgement it is part of legal practice. Hence, its truthfulness corresponds to truth₂. In this normative context, truth₂ is the judgment to which the authorised legal experts would have conceded if the judgement had been decided by them and not by $\mathcal{F}$. Moreover, since np is part of the legal practice, its explanation should be based on the normative model CMSLP. According to CMSLP, the truthfulness of np is grounded on the hierarchy HP_2 : there are propositions and norms situated lower than np in the hierarchy HP_2 such that np is derived from them via subsumption-deduction. Consequently, an explanation of np should consist of the subsumptive deductive arguments whose conclusion is np . Finally, the user whose data are filtered knows the normative explanation of output when they know those subsumptive-deductive arguments

3. The Role of Professional Norms in the Governance of Artificial Intelligence

The development, deployment, and use of artificial intelligence (AI) systems and AI-based technologies are governed by an increasingly complex set of legal, ethical, social, and other types of norms, which surface from various sources and contribute to what might be described as a patchwork of norms. Among this complicated landscape of modes of governance, this chapter zeroes in on the extent to which professional norms — and specifically norms in the development phase as expressed in formal documents such as code of ethics and ethical principles — may serve as a reservoir of norms and accountability mechanisms to include within the existing governance toolbox.

It explores the interface between AI and “the profession,” with an emphasis on new institutional arrangements and sources of norms that arise within the profession, such as corporate principles and employee demands. This chapter discusses trends of this fluctuating ecosystem through a suggested analytical framework for thinking about these professional norms of AI development within the broader context of the AI lifecycle, and hypothesizing about the future possibilities of professional norms within discussions of AI governance.

3.1 AI, Professions, and Norms:

Examining professional norms and ethics as a potential source and mode of governance of AI triggers fundamental questions about definitions and concepts: What is AI? What are the professions in general and in the context of AI specifically? What do we mean by professional ethics? No universally agreed upon definitions of these terms exist, and each of these concepts is itself in a state of flux. The meaning of AI remains amorphous, at least from a multidisciplinary perspective, as methods and techniques evolve and contexts of application change. The very notion of what a profession is and what professionalism stands for has shifted dramatically over time, particularly in knowledge economies.¹ And identifying and understanding the ethical questions to be addressed at the intersection of professions and AI is also a work in progress. Layered on top of each other, these three concepts – the profession, professional norms, and AI – create what one might describe as a perfect definitional storm, with an extraordinarily rich history, theory, and practices at its massive eye. In light of this complexity and uncertainty, this chapter takes a modest, pragmatic approach and offers selected observations based on the work of the authors in the context of a larger research effort on the ethics and governance of AI. The following initial triangulation of the core elements at play – AI, professions, and professional norms – frames our subsequent observations. One of the few areas of consensus among scholars is that there is no universally agreed upon definition for AI. The working definition we use in this chapter, however, encapsulates the complexity and contextuality of AI, including its history and future trajectory, the interdisciplinary stakeholders and researchers involved in AI. For present purposes, more important than the definition itself are some of the characteristics of AI systems, including their increased pervasiveness and impact on human autonomy and agency. Many reports document how AI-based technologies increasingly penetrate areas such as transportation, health, education, justice, news and entertainment, and commerce, to name just a few areas of life. The increased pervasiveness highlights the importance of the context for which AI systems are developed and in which they are embedded when considering normative questions and the role of professional norms aimed at addressing them. Perhaps the most fundamental attribute of many AI systems, however, is the varying degrees to which they affect human autonomy and shift it away from human beings towards machines, with potentially deep consequences also for concepts such as the profession and professionalism. Many of the changes associated with AI that will shape the nature of the profession can be seen as amplifications of tectonic shifts that have been going on for some time: Over the past few decades, a large body of multidisciplinary research has

documented and analyzed the transformations of the concepts of the profession and professionalism, including the seemingly paradoxical erosion of traditional liberal professions (e.g., lawyers and doctors) on the one hand and the growing appeal of these concepts across many other occupations on the other. The expanding application of “professions” across occupational groups, contexts, and social systems has a direct effect on the extent to which the concept of the profession – and its set of norms and enforcement regimes – has the potential to play a role in the governance of AI. This expansion suggests that professions and associated norms might be increasingly relevant as governance mechanisms of AI, particularly when it comes to the use of AI systems by professionals in different social contexts. The ideal example is a physician who uses an AI-based system to diagnose a patient’s disease, or a judge who relies on predictive analytics when making decisions about bail or sentencing. In these instances, the professional norms of ethics have the potential to help govern the use of the respective AI-based systems. Perhaps even more importantly, the conceptual shift away from traditional or “pure” professions towards what one scholar calls “mixed-up” and other forms of professionalism⁵ opens the door to examine how new and evolving non-traditional occupations existing alongside professions engaged in the development of AI can help fill the reservoir of norms and enrich the governance toolkit. This particular perspective, with focus on professions that are on the development side of AI, will be the focus for the remainder of this chapter. As notions of professions and professionalism are changing, the normative and institutional arrangements are shifting, too. In the case of traditional professions, control over professional practices was exercised through norms created and administered by professional associations. In recent times, not only have professional associations proliferated in concert with the professionalization of various occupations, but alternative institutional arrangements have emerged that shape the evolutionary path of professional norms and the ways in which they are enforced among professionals. This latter trend also pervades when looking at professional norms and ethics in the context of AI, where various non-traditional players – including NGOs and companies – are engaged in norm creation and, to a lesser extent, administration. Despite changes in the institutional set-up and governance of professional norms, the types of issues addressed by professional norms have remained largely stable by focusing on the regulation of the behaviour of the professionals themselves and the impact of their products and services on society. Similarly, many of the ethical questions about norms of the profession have remained relatively stable despite the broadening of these concepts. In this respect, AI might ultimately be a driver of deeper changes: professional norms and ethics in the age of AI might not only address the traditional questions, but also the future effects of the

increasingly autonomous systems that the profession creates – including feedback effects on the profession itself.

AI as Profession(s)?

AI-based technologies are often highly complex systems that require the collaboration of people with various types of knowledge, skills, and judgment across the different phases of AI system creation – including design, development, testing. Consider autonomous vehicles. These cars need designers, computer scientists, engineers, software developers, policymakers, legal experts, business representatives, among others, who work together to conceptualize and build self-driving cars and bring them to the street. Depending on geography, culture, and context, several of these activities involved in the creation of AI systems might be carried out by people that not only represent a discipline, but also belong to an occupation – some of which (self-)identify as a profession. For instance, many people working on the technical side of AI development are members of professional organizations such as The Institute of Electrical and Electronics Engineers (IEEE). However, determining what types of activities fall under a specific profession is not always straightforward. Seemingly well-defined activities like “engineering” are illustrative: An in-depth analysis suggests the lines between engineering activities, disciplines, occupations, and professions are blurry and change over time, particularly in light of changing socio-economic circumstances and technologies. This boundary-drawing challenge is further complicated when considering that individuals outside of established disciplines, occupations, and profession might also be involved in the development of AI-based technologies. As one scholar says, AI development work “can be done by those working entirely outside the framework of any professional accreditation” – and potentially even by amateurs.

Norms of the AI Profession(s)

The observations above suggest that the application of the concept of the profession and professionalism in the context of AI are still in flux. Further complicating the scope of professionalism with AI, the professional norms themselves are situated within various contexts and they interact with other explicit and implicit sources of norms. In corporate settings, for instance, these norms may also interact with extant frameworks for normative business ethics. The trends sketched above simultaneously build upon well-established ground and familiar territory (e.g., the earlier case of engineering), and are also more novel in terms of the intricate assemblage of activities, disciplines, occupations, and professions involved in the development of (typically) complex AI systems. The current ambiguity and diversity of possible perspectives is also reflected

when looking for norms that might be relevant at the nexus of AI and the professions. Two characteristics of the current norms landscape seem particularly relevant: Given the mélange of (quasi-)professional activities and actors involved, it is not surprising that the sources of norms transcend the set-up of professional associations that played the decisive role in the context of the traditional professions. Furthermore, the dynamic nature of the state of play suggests that relevant norms come in gestalt of “code of ethics,” but in other cases and depending on context might be less structured and more informal, at times even implicit in the form of normative routines of “everyday professional life.”

4. Teaching Machines to be Moral

For years, experts have warned against the unanticipated effects of general artificial intelligence (AI) on society. Ray Kurzweil predicts that by 2029 intelligent machines will be able to outsmart human beings. Stephen Hawking argues that “once humans develop full AI, it will take off on its own and redesign itself at an ever-increasing rate.” Elon Musk warns that AI may constitute a “fundamental risk to the existence of human civilization.”

More often than not, these dystopian prophecies have been met with calls for a more ethical implementation of AI systems; that somehow engineers should imbue autonomous systems with a sense of ethics. According to some AI experts, our future robot overlords can be taught right from wrong, akin to a good Samaritan that will always act justly on its own and help humans in distress.

Although this future is still decades away, today’s narrow AI applications need to be imbued with better moral decision-making. This is particularly necessary when algorithms make decisions about who gets access to loans, who gets promoted at work, or when self-driving cars have to calculate the value of a human life in hazardous situations.

Teaching morality to machines is hard because humans can’t objectively convey morality in a way that makes it easy for a computer to process. In moral dilemmas, humans tend to rely on gut feeling instead of cost-benefit calculations. Machines, on the other hand, need objective metrics that can be clearly measured and optimized. For example, an AI machine can excel in a game with clear rules and boundaries. Alphabet’s Deep Mind was able to beat the best human player of Go. Meanwhile,

OpenAI amassed “lifetimes” of experiences to beat the best human players at this year’s Dota 2 tournament, a video game competition.

But in real-life situations, optimization problems are vastly more complex. For example, how can a machine be taught to algorithmically maximise fairness or to overcome racial and gender biases in its training data? A machine cannot be taught what is fair unless the human teaching it has a precise conception of what fairness is.

This has led some authors to worry that a naïve application of algorithms to everyday problems could amplify structural discrimination and reproduce biases in the data on which they are based. In the worst case scenarios, algorithms could deny services to minorities impede employment opportunities or get the wrong political candidate elected.

Based on our experiences in machine learning, there are three ways to make machines more ethical.

First, AI researchers and ethicists need to formulate ethical values as quantifiable parameters. In other words, they need to provide machines with explicit answers and decision rules to any potential ethical dilemmas it might encounter. This would require that humans agree amongst themselves on the most ethical course of action in any given situation—a challenging but not impossible task. For example, Germany’s Ethics Commission on Automated and Connected Driving has recommended that automated cars be programmed to prioritize the protection of human life above all else. In the event of an unavoidable accident, the car should be “prohibited to offset victims against one another.” In other words, a car shouldn’t be able to choose whether to kill one person or many in an unavoidable crash situation.

Second, researchers need to collect enough data on explicit ethical measures to appropriately train AI algorithms. Getting appropriate data is challenging, because ethical norms cannot be always clearly standardised. Different situations require different ethical approaches, and some ethical dilemmas may not have a single

best course of action. One way of solving this would be to crowd source potential solutions to moral dilemmas from millions of humans. For instance, MIT's Moral Machine project shows how crowd sourced data can be used to train machines to make moral decisions in the context of self-driving cars.

Third, policymakers need to implement guidelines that make AI decisions with respect to ethics more transparent. Simply blaming an algorithm for making a mistake is not acceptable, but demanding full algorithmic transparency may also be technically untenable, because neural networks are simply too complex to be scrutinised by human inspectors. Instead, there should be more transparency on how engineers quantified ethical values before programming them, as well as the outcomes that the AI has produced as a result of these choices. For self-driving cars, for instance, this could imply that detailed logs of all automated decisions are kept at all times to ensure their ethical accountability.

Failing to imbue ethics into an AI system allows an algorithm to decide what's best for us. For example, in an unavoidable accident situation, the self-driving car needs to make a decision. But if the car's designers fail to specify a set of ethical values that could act as decision guides, the AI system may come up with a solution that causes more harm.

Machines cannot be assumed to be inherently capable of behaving morally. Humans must teach them what morality is, how it can be measured and optimized. For AI engineers, this may seem like a daunting task. After all, defining moral values is a challenge mankind has struggled with throughout its history. Nevertheless, the state of AI research requires engineers and ethicists to define morality and quantify it in explicit terms. Engineers cannot build a "Good Samaritan AI" as long as they lack a formula for the Good Samaritan human.